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Crypto Data Tools: What You Need to Know



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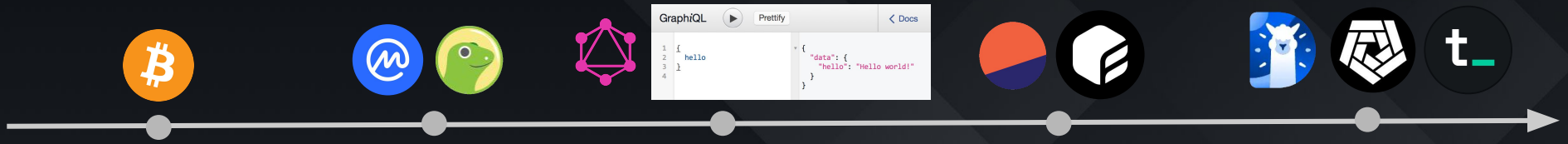
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Evolution of Crypto Data



2010

Block Explorer becomes the first tool allowing to view the details of transactions on the Bitcoin network

2012 – 2014

Crypto asset data aggregators emerge showing coin prices and trading volumes

2015

GraphQL made publicly open-sourced, allowing support for smart contracts and DeFi data

2017 – 2018

The launch of SQL query-friendly platforms, such as Dune Analytics and Flipside. Data analysts can now easily query on-chain transactions

2020 – Now

Comprehensive data monitoring platforms and AI-powered tools such as DeFillama and Arkham Intelligence emerge

Block explorers, such as Etherscan, serve as basic data analytics providers, enabling users to examine raw on-chain data. However, getting insight into a protocol's daily active users, revenue, and other important variables is not straightforward with such tools.

On-chain data analysis is important because it helps to transform raw transaction data into valuable insights, enabling effective evaluation of crypto projects or sectors and promoting informed decision-making and strategic planning. The need for comprehensive blockchain data analytics led to the rise of more sophisticated data providers.

The soaring demand for **artificial intelligence (AI)-driven blockchain analytics** stems from this technology's ability to efficiently identify **fraudulent activities** and **enhance trading strategies**. Blockchain analytics companies, such as Arkham Intelligence, harness AI to streamline blockchain transaction monitoring and data analysis, delivering valuable insights to users.

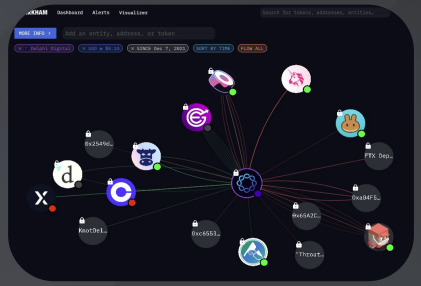
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Example Use Cases

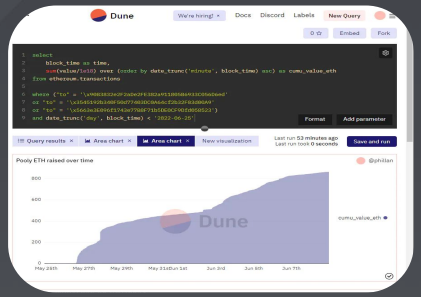
Tracking Funds' Flows

Identifying **fund movements** between various entities originating from the primary entity and monitoring "whale" addresses with substantial asset balances. By **leveraging AI technologies** such as that of Arkham Intelligence, one can obtain more comprehensive insights and linkages.

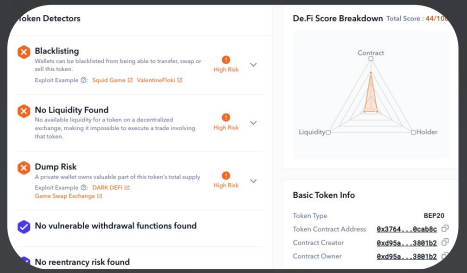


Developing Customized Metrics

Developing tailored metrics for specific needs, or those needed to better understand contracts and transactions data, through the use of **SQL queries** on platforms such as Dune Analytics or Flipside Crypto.

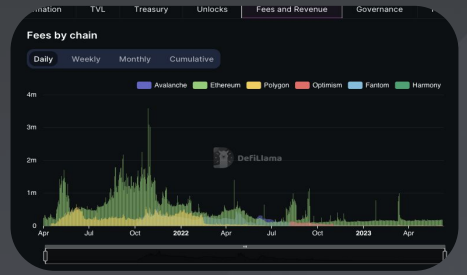


Auditing Smart Contracts



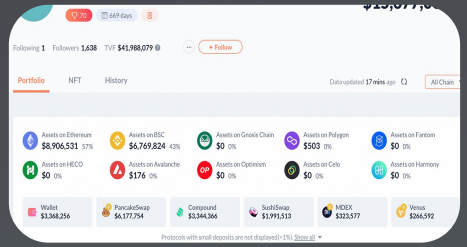
Analyzing smart contracts to **improve security and reduce vulnerabilities, preventing potential exploitation.** De.Fi is an example of a tool that introduces risk factor scoring.

Analyzing Projects' Fundamentals



Understanding crypto projects' fundamental metrics plays a vital role in making informed investment decisions. For example, for decentralized exchanges (DEXs), in particular AMM DEXs, TVL is a crucial metric influencing bid/ask spreads for efficient trades.

Copy Trading



Monitoring large wallet holders and professional traders' transactions via wallet explorers like DeBank, and mimicking their strategies to increase chances of achieving sizeable returns.

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Crypto Data Tools Map

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About the Crypto Data Tools Map

The Crypto Data Tools Map provides an overview of the available tools and infrastructure projects in the crypto ecosystem that can be used in the analysis of digital asset-related data.

We have identified a variety of relevant data types, encompassing both on-chain and off-chain sources, that go beyond the most commonly used transaction data. However, note that some of the data types are not yet readily available in a user-friendly format via data tools and providers.

This map is compiled based on publicly available information. It is not intended to present an exhaustive list of projects in the crypto data space. Our intention is to present a balanced mix of established players and emerging projects, so certain tools or projects may or may not have been included. Projects showcased are mapped according to their primary use cases. For suggestions, please provide your feedback here.

The mention of specific projects does not constitute endorsement or recommendation by Binance.

Data Tools and Infrastructures
Vertical spotlights

Landscape Monitoring Data Tools

These tools are designed to provide a comprehensive overview of the market landscape, encompassing a wide array of variables across CeFi and DeFi as well as insights derived from social listening.

Contract & Transaction Analysis Tools

This family of tools mainly offers dashboarding features and contract risk assessment capabilities. They enable individuals to perform an in-depth analysis of smart contracts and transactions, transforming raw data into valuable insights.

Data-Focused Infrastructures

These projects serve as the foundation for enabling a wide range of use cases within Web3 DApps. They empower the democratization of data, for example, by facilitating the integration of oracles that provide precise market data to DeFi protocols.

Types of on-chain data



Transaction

- Buying/Selling/Swapping
- Transferring/Donating
- Borrowing/Lending
- Staking/Unstaking/Mining



Social Graph

- Friending/Unfriending
- Following/Unfollowing
- Referring/Getting referred
- Liking/Unliking



DAO Governance

- Voting for
- Proposing
- Voting against
- Abstaining



Collectibles

- Minting/Creating NFT
- Curating exhibition
- Burning
- Auctioning



Certification

- Event attendance (POAP)
- Educational achievements
- Project contributions
- Digital identity



Gaming

- Winning/Losing
- Participation
- Teaming up
- Earning points



Others

- Tokenized Real-World Assets
- Music NFTs
- Carbon Credits

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Landscape Monitoring Data Tools

Centralized Exchanges Market Data

THE BLOCK · MESSARI · KAIKO · coinglass · intotheblock · glassnode · CryptoQuant

DeFi

APY.vision · DEXTools · DefiLlama · DappRadar · MESSARI · dPrism · dex.guru · intotheblock · nansen · DEXSCREENER

Entity Trackers

ARKHAM · nansen

Fundraising Trackers

THE BLOCK · MESSARI · ROOTDATA · DefiLlama

Token Discovery

UNIBOT · MAESTRO

NFT

DefiLlama · CRYPTO SLAM! · NFTSCAN · DappRadar · nansen · intotheblock · icy.tools · dPrism · NFTGO

DAO

MESSARI · snapshot · boardroom

Social Analytics

KAITO · santiment · Yap · TwitterScan · LunarCrush

Macro

CryptoQuant · intotheblock · glassnode · THE BLOCK · DefiLlama · L2BEAT · MESSARI · KAIKO · The Tie

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Landscape Monitoring

Centralized Exchanges Market Data

Project	Paid/Free	Description
KAIKO	Paid	Kaiko predominantly offers data from centralized exchanges, such as order book data.
glassnode	Paid/Free	Glassnode is a provider of both on-chain and centralized exchange data.
CryptoQuant	Paid/Free	CryptoQuant is a on-chain, centralized market and signals data provider for trading-focused stakeholders.

DeFi

Project	Paid/Free	Description
DefiLlama	Free	Defillama tracks key metrics like revenue and TVL for different DeFi projects from various sectors (Lending, DEX, Perp).
DEXSCREENER	Free	Dex Screener is a DEX-focused platform that shows trending and new liquidity pools for liquidity farmers.
nansen	Paid	Nansen is an on-chain monitoring platform for crypto investors, with *DeFi Paradise* as a core DeFi analytics offering.

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Landscape Monitoring

Entity

Project	Paid/Free	Description
ARKHAM	Free (Beta)	Arkham Intelligence provides detailed data on entities using proprietary AI and off-chain intelligence to to deanonymize addresses.
nansen	Paid	Nansen is an on-chain monitoring platform for crypto investors, with *Entity Profiler* as a core analytics offering.

NFT

Project	Paid/Free	Description
CRYPTO SLAM!	Free	CryptoSlam offers insights into trending NFT collections and latest NFT sales.
icy.tools	Paid/Free	Icy Tools is an NFT analytics platform that offers real-time sales, floor price, and volume data for NFT traders.
	Paid/Free	Rarity Sniper delivers NFT analytics with a focus on NFT traits and allows users to track NFT minting events.

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Landscape Monitoring

DAO

Project	Paid/Free	Description
MESSARI	Paid/Free	Messari provides basic DAO proposal tracking and advanced metrics for paid plan tiers.
boardroom	Free	Boardroom is a dashboard built for active DAO participants to track latest proposals.
snapshot	Free	Snapshot is an off-chain voting platform that allows DAOs, DeFi protocols, and NFT communities to participate.

Social analytics

Project	Paid/Free	Description
KAITO	Paid	Kaito is a platform storing unstructured social media data that users can query.
LunarCrush	Paid/Free	LunarCrush offers insights from various social media outlets.
TwitterScan	Free	TwitterScan provides insights into various trending crypto-related topics based on the analysis of Twitter activity.

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Landscape Monitoring

Macro

Project	Paid/Free	Description
L2BEAT	Free	L2Beat is a platform that provides chain-focused metrics for various layer-2 projects.
MESSARI	Paid/Free	Messari offers an overview of high-level metrics for the crypto market.
THE BLOCK	Paid/Free	The Block is a Web3 news outlet and research and data provider offering macro-level insights.

Fundraising

Project	Paid/Free	Description
ROOTDATA	Free	Rootdata is a platform that aggregates the latest fundraising information for Web3 projects.
MESSARI	Paid	Messari provides the data on the latest Web3 fundraising rounds.
DefiLlama	Free	Defillama offers information on Web3 fundraising.

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Landscape Monitoring

Token Discovery

Project	Paid/Free	Description
UNIBOT	Paid/Free	Unibot provides real-time alerts on newly created tokens, as well as sniping services.
MAESTRO	Paid/Free	Maestro is a token discovery and trading tool.

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Contracts & Transactional Analysis Tools

Overview

Dashboard Platforms



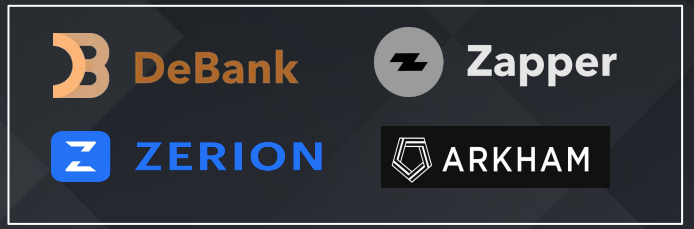
Investigative



Block Explorers



Wallet Explorers



Contract Audit Tools



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Contracts & Transactional Analysis

Dashboard Platforms

Project	Paid/Free	Description
Covalent	Free	Covalent offers a dashboarding query tool called Increment and supports over 100 chains.
Dune	Paid/Free	Dune is a community-focused platform that allows users to query blockchain data and build shareable dashboards.
flipsidecrypto	Free	Flipside Crypto allows users to query blockchain data and participate in data bounties for rewards.

Block Explorers

Project	Paid/Free	Description
Etherscan	Free	Etherscan is a block explorer and analytics platform for Ethereum whose team is also behind products like BscScan and Snowtrace.
Blockscout	Free	Blockscout is an explorer-as-a-service platform focused on EVM chains that offers support for more than 20 networks.
BLOCKCHAIR	Free	BlockChair provides block explorer services and API for EVM and UTXO chains.

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Contract & Transaction Analysis

Contract Auditing Tools

Project	Paid/Free	Description
bscheck.eu	Free	Bscheck provides audits of token contracts on five supported chains.
Cyberscope	Free	Cyberscope offers the auditing of smart contracts on EVM chains for issues like malicious code.
	Free	De.Fi offers smart contract audits with scoring for various risk factors.

Investigative Tools

Project	Paid/Free	Description
bubblemaps	Paid/Free	Bubblemaps is a data visualization platform that makes the analysis of transactional flows more accessible and convenient for users.
breadcrumbs	Free/Paid	Breadcrumbs provides visualisations that help track transaction flows between addresses or entity clusters as well as transaction monitoring.
ARKHAM	Free (Beta)	Arkham Intelligence provides detailed insights into fund flows between addresses and entities.

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Contracts & Transactional Analysis

Wallet Explorers

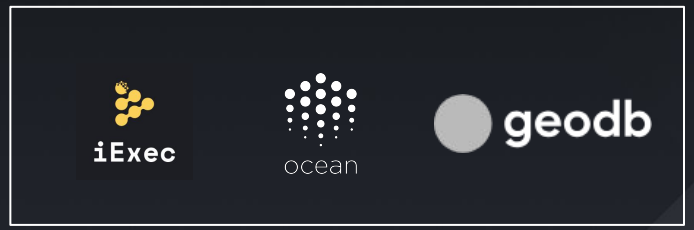
Project	Paid/Free	Description
ZERION	Free	Zerion is a non-custodial wallet provider that also offers a web-based portfolio management tool with tracking of other wallets.
Zapper	Free	Zapper offers a wallet portfolio management tool for tokens and NFTs as well as tracking of other tagged wallets.
DeBank	Free	Debank is a platform that offers wallet portfolio tracker for EVM chains.

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Data Infrastructure
Overview











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Data Infrastructure

Storage

	Filecoin is an open-source cloud storage protocol, marketplace, and incentive layer.
	Arweave offers permanent and decentralized storage in an open ledger.
STORJ	Storj offers low-cost cloud storage with zero-trust security.

Data Marketplaces

ocean	Ocean Protocol is a decentralized data market that allows users to publish, discover, and consume data for AI modelling purposes.
geodb	GeoGB is a decentralized data platform that aims to democratize the big data market.
iExec	iExec is a decentralized cloud platform that allows users to rent out computing assets like datasets.

Database Solutions

ceramic	Ceramic is decentralized data network that powers an ecosystem of interoperable Web3 applications and services.
TABLELAND	Tableland is a Web3-native SQLite database that allows users to read and write data from any app or smart contract.
kwil	Kwil is a permissionless relational database designed for DApps and decentralized platforms.

Data Indexing

the graph	The Graph is an indexing protocol for querying networks like Ethereum.
0xScope	0xScope is a protocol that aims to standardize Web2 & Web3 data by building a knowledge graph.
Covalent	Covalent is a chain-indexing network with 100+ supported chains.

Oracles

Chainlink	Chainlink is a decentralized oracle network providing tamper-proof inputs, outputs, and computations to support smart contracts.
PYTH	Pyth delivers real-time on-chain market data.
API3	API3 is a first-party oracle network.

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About Binance Research

Binance Research is the research arm of Binance, the world's leading digital-asset infrastructure provider and ecosystem. The team is committed to delivering objective, independent, and comprehensive analysis and aims to be the thought leader in the crypto space. Our analysts publish insightful thought pieces on topics related, but not limited to, the crypto ecosystem, blockchain technology, and the latest market trends.



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Contributors



Keng Ying Sim

Keng Ying (“Sky”) is currently a Research Data Analyst at Binance. Before joining Binance, she worked as a Research Analyst at Bybit and Citi, the latter with a focus on offshore wealth management analytics.

Sky is an on-chain data enthusiast who enjoys going down the rabbit hole in her free time.



Nicholas Tan Yi Da

Nicholas is currently a Research Data Analyst intern on Binance’s Research Data Team. He is a recent Engineering Systems and Design graduate of the Singapore University of Technology and Design.

He is passionate about data analytics, particularly in areas like DeFi and RWAs.

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